

Progress Report Sprint 4

- Course Name: CS461 Senior Software Engineering Project I
- Date: 11/23/25
- Team Name: TaxonBodyMassML
- Team ID: CS.032
- Student Names and Emails:
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Summary

- Updated UI according to design recommendations from our project partner
- Made CSS styling responsive to screen size
- Created decision tree prototype as a proof of concept for the random forest

Progress vs Plan

PLANNED ITEM	STATUS (% OR DONE/PARTIAL/NOT DONE)	LINKS (ISSUE/PR)	NOTES/BLOCKERS	PLAN CHANGE
CSS STYLING	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/web-dev/web_dev	Did research, but focused on responsiveness over cosmetics.	Work toward styling after everything else is clean
PREDICTIVE ALGS	Partial	https://github.com/marknovak/TaxonBodyMassML/commit/3b5f58fffe71f521b50bc4adb1dc2720e9a70d9	Created basic decision tree prototype. However, it cannot be finished until the data is done being merged.	
DATA CLEANING	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning	Standardized methods for removing missing data.	
DATA MERGING	Partial	https://github.com/marknovak/TaxonBodyMassML/tree/data-combination	Need to figure out how to optimize the data merging process because of the large dataset and need to figure out how to deal with rows with	We are prioritizing this as the main focus of the next sprint.

Evidence of Working Software

Web Development latest update commit hash: 23ceef37e0c3a9a1a1c1ca0fbb398aeb8440f48f

Link to UI: https://praterh.github.io/HaileysTaxonBodyMassML/web_dev/index.html

Video of Responsive CSS:  Screencast from 2025-11-23 15-40-48.webm

Risk and Quality

- After having an outside party view the web application, it was clear the CSS responsible for the design layout needed to be tweaked. Because the elements had fixed positions, anyone with a smaller screen than mine would have a much worse experience interacting with the UI.
 - Owner: Hailey
 - Status: Complete
 - Concrete actions: I went back to my CSS and added another section containing a media query. This section would implement entirely different layout styling for screens smaller than 768 pixels. I tested it by resizing my browser window repeatedly to make sure the experience remained consistent despite screen size.
- Across various parts of the code base, there were several parts of the data management and manipulation in which there were different procedures for handling missing rows in the dataset. This can cause inconsistencies which may later be problematic.
 - Owner: Grant
 - Status: Complete
 - Concrete Actions: I went through and standardized the parts of the code base for which there were inconsistencies. The procedure is only to drop the columns with missing values for the “taxon” and “mass_g” variables. The other variables are not crucial to the project, and therefore are optional.

Next Goals

- Begin work on the UI’s help page; this will work toward REQ-004, allowing novice users to interact with the application easily via an information tab
- Create a functional decision tree model as a proof of concept for the random forest. This corresponds with REQ-007 because a random forest can be clearly visualized, so it will aid in transparency.
- Complete the data merging process so that we can start the predictive modeling process. This is necessary for the completion of REQ-002, which is the mass prediction requirement.

Individual Contributions

Grant:

- Working on branches: <https://github.com/marknovak/TaxonBodyMassML/tree/random-forest>, <https://github.com/marknovak/TaxonBodyMassML/tree/data-partitioning>
- Updated both prototype lookup and data visualization to comply with PR review suggestions
- Created high-level architecture diagram
- Set up basic decision tree prototype (still needs some updates before full functionality)

Hailey:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/web-dev>
- Incorporated suggestions from project partner into javascript movements and page layout
- Made the css-styling more responsive and small-screen compatible
- Researched effective color schemes for scientific web applications

Anish:

- Working on branch: <https://github.com/marknovak/TaxonBodyMassML/tree/data-combination> and <https://github.com/marknovak/TaxonBodyMassML/tree/test-cases>
- Worked on data merging of the files
- Trying to optimize time taken to merge files
- Deleted any corrupted data(multi-data type cells and merged cells from different columns)

Team Reflection

This week was more difficult for us than the last; we didn't make as much progress as we would have liked on any front. As we have approached the end of the term, we have grown progressively busier and recognize that we haven't put as much time into our project as we should have.

However, the progress we did make felt good to accomplish. The web design layout is becoming more and more complete, with only the styling and some written content holding it back from being a complete application. We also standardized our data cleaning techniques across our codebase, allowing us the chance to begin working on our algorithm without annoyances from data that isn't clean and consistent.

The highlight of this sprint was the meeting with our project partner. During this meeting, we got good feedback from him on the UI and asked for his opinions on the two different approaches we were considering for our ML component.